

Discrete Mathematics Assignment 4

Due Date: April 24, 2026

Write clear and concise handwritten solutions. You must also upload a soft copy of it on the moodle page. It is fine to discuss with others, but your solutions must be in your own words that you have fully understood.

1. Prove that the total number of antichains in the power set poset $(2^{[n]}, \subseteq)$ is bounded by $(n+1)^{\binom{n}{2}}$.
2. Let $\mathcal{F} \subset \binom{[n]}{k}$ be a family of k -subsets of $[n]$ such that any $k+1$ subsets $A_1, A_2, \dots, A_{k+1} \in \mathcal{F}$ intersect. That is, $\bigcap_{i=1}^{k+1} A_i \neq \emptyset$. Then show that all of them must intersect: $\bigcap_{A \in \mathcal{F}} A \neq \emptyset$.
3. Prove that a graph $G = (V, E)$ is bipartite if and only if it has no cycles of odd length.
4. Let $G = (U_1, U_2, E)$ be a bipartite graph such that for every subset $S \subseteq U_1$ we have $|N(S)| \geq |S| - d$, where $N(S) \subseteq U_2$ is the neighborhood of S and d is a fixed positive integer. Show that G has a matching of size $|U_1| - d$.